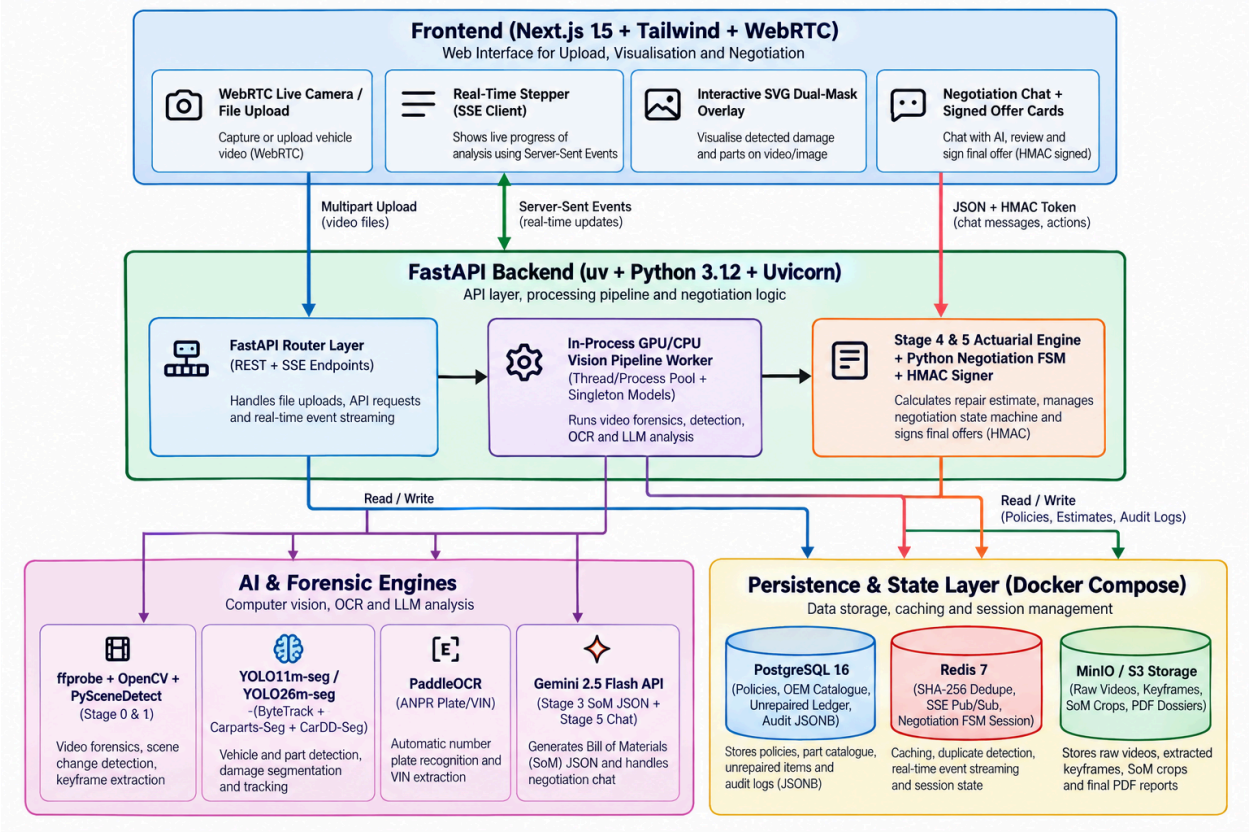
TECH STACK



Layer	Chosen Stack	Industry Alternative	Primary Use Case in Auto-Settle AI	Architecture & Constraints
Python & Package Manager	**uv + Python 3.12**	Poetry / Conda / pip	10–100x faster dependency resolution, deterministic uv.lock, unified virtual	Deterministic locks for reproducible backend pipeline worker builds.

			environment management.	
API & ASGI Server	**FastAPI + Uvicorn (uvloop)**	Granian / Gunicorn + Uvicorn	Async REST API, Server-Sent Events (SSE) for real-time stage progress, Pydantic v2 validation.	API router handles multipart video uploads and streams real-time SSE updates to clients.
Relational Database	**PostgreSQL 16 (via Docker/Supabase)**	SQLite (dev only) / MySQL	Stores insured_policies, oem_parts_catalogue, unrepaired_claims_ledger, and ACID claim state transitions.	Docker-managed persistence layer storing policies, estimates, and JSONB audit logs.
ORM & Migrations	**SQLAlchemy 2.0 (Async) + Pydantic v2**	SQLModel / Prisma	Strict type-safe database queries for Stage 0 policy checks and Stage 4 OEM pricing lookups.	Async ORM queries executed by backend API & Actuarial engines.
Cache, Locks & Pub/Sub	**Redis 7 / Valkey (Docker)**	In-Memory Dict (single-worker only)	Stores Stage 5 Negotiation FSM state, SHA-256 video idempotency locks, and SSE	Pub/Sub broker enabling real-time streaming between worker and frontend.

			pipeline progress events.	
Object Storage	**MinIO (Local S3) or Cloudflare R2**	AWS S3 / local disk	Stores raw walkaround videos, extracted best keyframes, Set-of-Marks (SoM) crops, and PDF audit trails.	Persistent blob storage connected across Vision Worker and Persistence layer.
Vision Segmentation	**Ultralytics YOLO11m-seg / YOLO26m-seg**	Mask R-CNN / SAM 2	Stage 1 ByteTrack continuity tracking and Stage 2 Dual-Mask Segmentation (Carparts-Seg + CarDD).	Runs in-process on Vision Pipeline Worker with ByteTrack and Dual-Mask segmentation.
Inference Engine	**ONNX Runtime (onnxruntime-gpu) / PyTorch FP16**	NVIDIA Triton / TensorRT	In-process Singleton GPU/CPU inference for dual YOLO models without microservice serialization overhead.	Thread/Process Pool with Singleton models avoiding microservice serialization bottlenecks.

Polygon & Geometry Math	**Shapely (GEOS C-library) + NumPy**	Pixel-array bitwise AND	Fast, exact vector polygon intersection (IoP) and 5% boundary spillover filtering in Stage 2.	In-process spatial analysis executed during Stage 2 processing.
Video & Forensic Tools	**ffprobe + OpenCV + PySceneDetect**	MoviePy / PyAV	Stage 0 container tag inspection (CapCut/Premiere), blur/luminance gates, and Stage 1 scene-cut detection.	Runs keyframe extraction and video forensics on initial upload ingress.
ANPR / License Plate OCR	**PaddleOCR (PP-OCRv4) + Gemini Fallback**	EasyOCR / Tesseract	Stage 1 license plate and VIN reading to ground vehicle identity against insured_policies.	Executed within Vision Engine pipeline with LLM fallback support.
VLM & Negotiation LLM	**Gemini 2.5 Flash (Structured JSON Mode)**	GPT-4o / Claude 3.5 Sonnet	Stage 3 SoM depth/rust/dirt verification (temperature=0.0) and Stage 5 bounded negotiation phrasing.	Generates Stage 3 SoM JSON and drives Stage 5 negotiation chat via external API calls.

Frontend UI	**Next.js (App Router) + Tailwind + Shadcn UI**	Vite + React / Streamlit	WebRTC live camera capture, interactive SVG segmentation mask overlays, and signed Offer Cards.	Client web interface for video capture, live stepper visualization, and HMAC-signed offer acceptance.

Detailed Technical Architecture & System Constraints

- **Frontend & User Interface:** Next.js 15 (App Router) + Tailwind CSS + WebRTC web interface for vehicle video upload, real-time stage progress monitoring via Server-Sent Events (SSE), interactive SVG dual-mask overlays, and an interactive negotiation chat interface with HMAC-signed final offer cards.
- **Backend Infrastructure & API Layer:** FastAPI backend built on Python 3.12 managed via uv and served with Uvicorn (uvloop). Houses FastAPI router endpoints (REST + SSE) and orchestrates data flow between workers and persistence layers.
- **In-Process Vision Pipeline Worker:** Utilizes Thread/Process Pools and Singleton GPU/CPU models to run video forensics (ffprobe, OpenCV, PySceneDetect), object/damage detection and tracking (YOLO11m-seg / YOLO26m-seg with ByteTrack and Carparts-Seg + CarDD), ANPR plate/VIN extraction (PaddleOCR PP-OCRv4), and Gemini 2.5 Flash Set-of-Marks (SoM) JSON analysis in-process without microservice serialization overhead.
- **Actuarial Engine & Negotiation FSM:** Python-based actuarial engine for Stage 4 repair cost estimation integrated with Stage 5 negotiation Finite State Machine (FSM) and an HMAC signer for securing final offers.
- **Persistence & State Layer (Docker Compose):** PostgreSQL 16 relational database for insured policies, OEM catalogue, unrepaired claims ledger, and JSONB audit logs; Redis 7 for SHA-256 video deduplication locks, real-time SSE Pub/Sub, and negotiation session state; MinIO / S3 Object Storage for raw videos, keyframes, SoM crops, and PDF reports.